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Phases Complete

7 of 7 phases.

Phase Resets

Phase 1: 0
Phase 2: 0
Phase 3: 0
Phase 4: 0
Phase 5: 0
Phase 6: 0
Phase 7: 0
Phase 8: 0

Popup Questions

Why is it important to stand downstream of your sampling device when testing for water quality measures?
Incorrect Attempts: 0

Why were two samples taken from each site on the river?
Incorrect Attempts: 0

Which chemical should be tested first based on the emailed reports from phase 1 regarding dead fish, strong chemical odors, and dog illness linked to time in the river?
Incorrect Attempts: 1

What additional recommended PFAS testing should be done based on these results?
Select all that apply.
Incorrect Attempts: 6

Lab setup

Identify emails with situations appropriate for DEQ investigation

Dead fish appearing in the river : Yes
Dog illness linked to time in river : Yes
Illegal poaching observed : No
Strong chemical odors outside my house : Yes
Too many potholes on my street : No

Analyzing handheld device results

Based on your collected data, which of the following average values from the 4 sampling locations fall within the normal range? Select all that apply.

Temperature
pH
Dissolved oxygen
Turbidity
Phosphorus
Nitrates
Fecal coliform

Based on the average values, are there any water quality values that are concerning?

No, because all values were within the normal ranges.

Apply what you have learned

Several samples of fish were caught by the department of natural resources in the vicinity of the cookware manufacturing plant and a sample was taken from a filet of each fish. The results are:

- Fish 1- 90,000 ppt
- Fish 2- 120,000 ppt
- Fish 3- 130,000 ppt

Why are the levels of PFAS in the fish so much higher than in the water?

PFAS chemicals can bioaccumulate in animals over time.

In what ways should the DEQ respond to the discovery of high PFAS levels in this area?
Select all that apply.

Advise residents in the area to avoid eating fish caught near this area of the river.
Institute additional testing near the cookware plant to search for additional contamination.
Offer educational seminars about PFAS pollution and its impacts.

Looking closer at the PFAS cycle in the environment, what are two other ways that PFAS could enter a river?
Select all that apply.

Groundwater release from a different factory
Wastewater treatment plant discharge into the river
Residential homes

Based on the results from this lab, would this be considered a point or nonpoint source of PFAS pollution?

This is a point source of pollution.

Collected Lab Data

	Site 1	Site 2	Site 3	Site 4	Avg.	Normal values
Temperature (°C)	18	19	19	17	18.3	14 - 22
pH	7.5	6.8	7.2	7.1	7.2	6.5 - 8.5
Dissolved O_2 (mg/L)	8	12	5	9	8.5	Above 6.5
Turbidity (NTU)	15	33	22	9	19.8	10 - 100
Phosphorus (P) (mg/L)	0.02	0.08	0.01	0.03	0.04	Less than 0.05
Nitrates (NO_3) (mg/L)	5	3	3	6	4.3	Less than 50
Fecal coliform (FC) (cfu per 100 mL)	5	4	5	9	5.8	Less than 60

Chemical	Site 1	Site 2	Site 3	Site 4
PFAS (ppt)	180	65	341	76